

Assignment 2

Student Name	Amit Panorekar	Roll Number	20968
Batch	SY. BSc Data Science	Topic	WB(Assignment-2)

Q1) Create a database named CollegeDB and a table named Student.

The table should contain the following columns:

Column Name	Data Type
StudentID	INT
StudentName	VARCHAR(50)
Age	INT
Course	VARCHAR(30)
City	VARCHAR(30)
Marks	INT

Add a primary key on StudentID and insert at least 10 records.

QUERY:

```
CREATE DATABASE CollegeDB;
```

```
USE CollegeDB;
```

```
CREATE TABLE Student (
```

```
    StudentID INT PRIMARY KEY,
```

```
    StudentName VARCHAR(50),
```

```
    Age INT,
```

```
    Course VARCHAR(30),
```

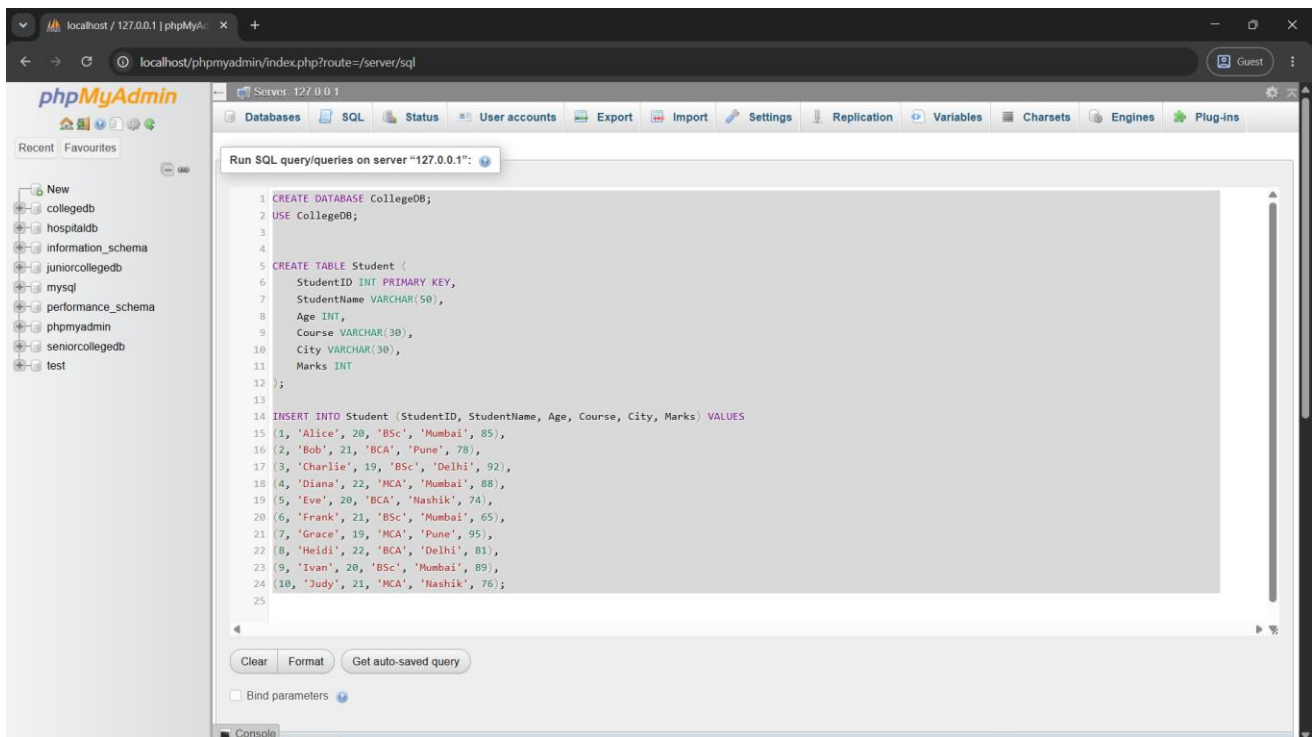
```
    City VARCHAR(30),
```

```
    Marks INT
```

```
);
```

```
INSERT INTO Student (StudentID, StudentName, Age, Course, City, Marks) VALUES
```

(1, 'Alice', 20, 'BSc', 'Mumbai', 85),
(2, 'Bob', 21, 'BCA', 'Pune', 78),
(3, 'Charlie', 19, 'BSc', 'Delhi', 92),
(4, 'Diana', 22, 'MCA', 'Mumbai', 88),
(5, 'Eve', 20, 'BCA', 'Nashik', 74),
(6, 'Frank', 21, 'BSc', 'Mumbai', 65),
(7, 'Grace', 19, 'MCA', 'Pune', 95),
(8, 'Heidi', 22, 'BCA', 'Delhi', 81),
(9, 'Ivan', 20, 'BSc', 'Mumbai', 89),
(10, 'Judy', 21, 'MCA', 'Nashik', 76);



The first screenshot shows the phpMyAdmin interface where a new database 'CollegeDB' is created and selected. Then, a table named 'Student' is created with the following schema: `CREATE TABLE Student ( StudentID INT PRIMARY KEY, StudentName VARCHAR(50), Age INT, Course VARCHAR(30), City VARCHAR(30), Marks INT );`. Finally, 10 rows of student data are inserted into the table.

The second screenshot shows the 'student' table in the 'Browse' view. It displays 10 rows of data with columns: StudentID, StudentName, Age, Course, City, and Marks. A tooltip indicates that the table is sortable and columns are clickable.

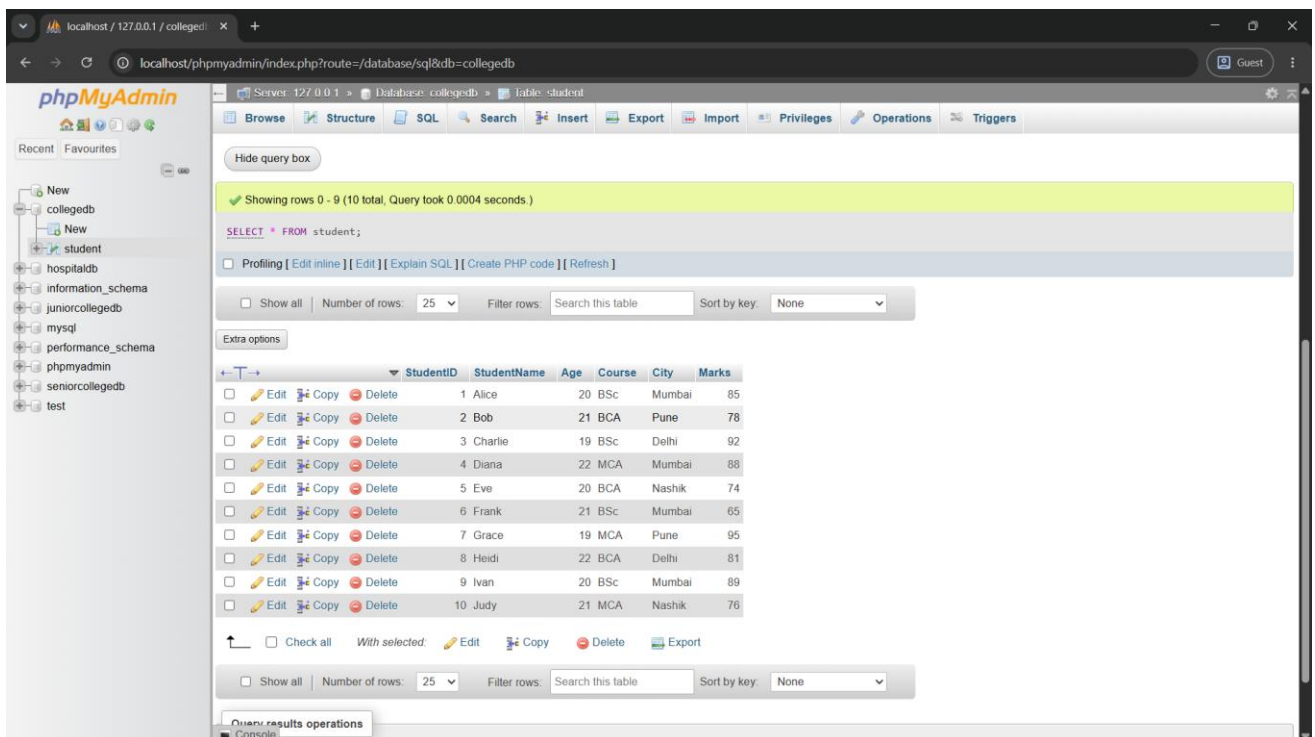
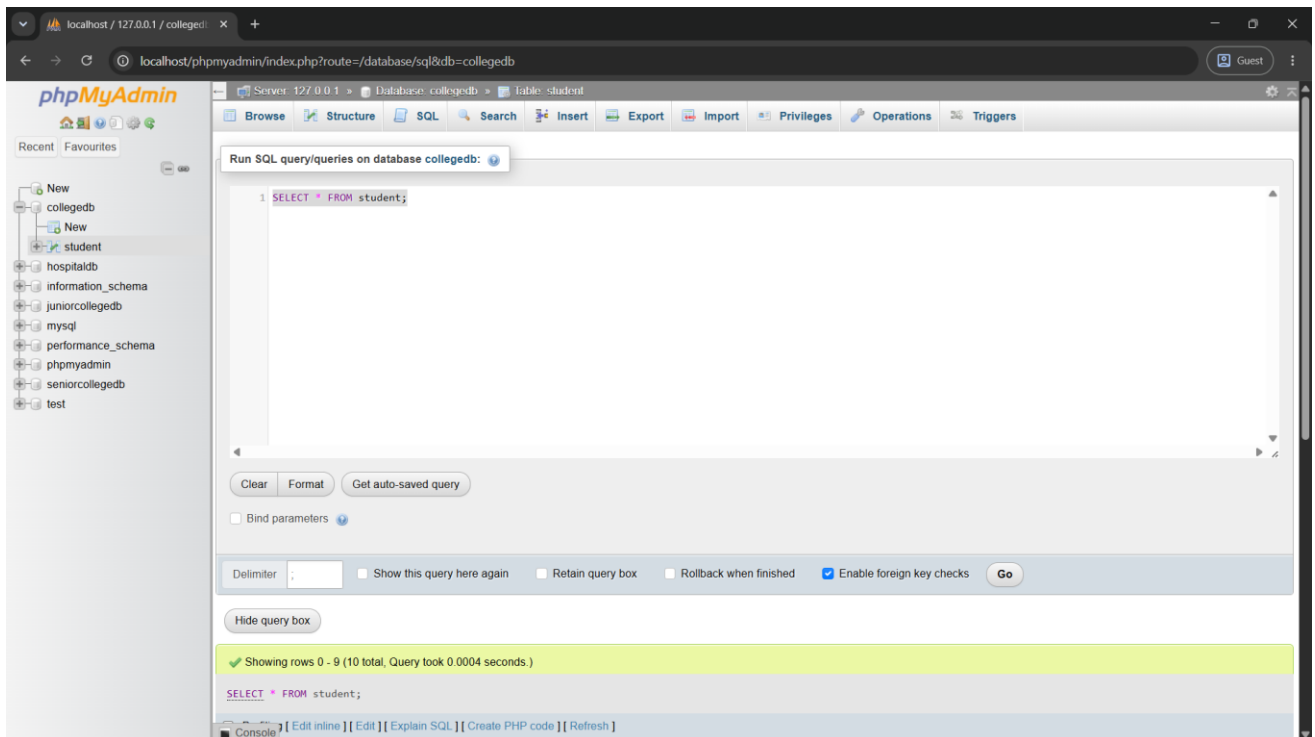
StudentID	StudentName	Age	Course	City	Marks
1	Alice	20	BSc	Mumbai	85
2	Bob	21	BCA	Pune	78
3	Charlie	19	BSc	Delhi	92
4	Diana	22	MCA	Mumbai	88
5	Eve	20	BCA	Nashik	74
6	Frank	21	BSc	Mumbai	65
7	Grace	19	MCA	Pune	95
8	Heidi	22	BCA	Delhi	81
9	Ivan	20	BSc	Mumbai	89
10	Judy	21	MCA	Nashik	76

Q2) Write a MySQL query to display all records from the Student table.

The output should display all columns and all rows.

QUERY:

`SELECT * FROM student;`



Q3) Write a MySQL query to display only the following columns:

1. StudentName
2. Course
3. City

QUERY:

`SELECT StudentName,Course,City FROM student;`

The first screenshot shows the phpMyAdmin interface with a SQL query entered in the query box:

```
SELECT StudentName, Course, City FROM student;
```

The second screenshot shows the results of the query, displaying a table with 10 rows of student data:

StudentName	Course	City
Alice	BSc	Mumbai
Bob	BCA	Pune
Charlie	BSc	Delhi
Diana	MCA	Mumbai
Eve	BCA	Nashik
Frank	BSc	Mumbai
Grace	MCA	Pune
Heidi	BCA	Delhi
Ivan	BSc	Mumbai
Judy	MCA	Nashik

Q4) Write a MySQL query to display students who belong to the city Mumbai. The output should include:

1. StudentID
2. StudentName
3. City
4. Marks

QUERY:

SELECT StudentID,StudentName,City,Marks

FROM student

WHERE City = 'Mumbai';

The first screenshot shows the phpMyAdmin interface with a SQL query entered in the query box:

```
1 SELECT StudentID,StudentName,City,Marks
2 FROM student
3 WHERE City = 'Mumbai';
4
```

The query is executed, and the results are displayed in a table with 4 rows (0-3) and 4 columns (StudentID, StudentName, City, Marks). The results are:

StudentID	StudentName	City	Marks
1	Alice	Mumbai	85
4	Diana	Mumbai	88
6	Frank	Mumbai	65
9	Ivan	Mumbai	89

The second screenshot shows the same phpMyAdmin interface, but the query results are displayed in a table with 4 rows (0-3) and 4 columns (StudentID, StudentName, City, Marks). The results are:

StudentID	StudentName	City	Marks
1	Alice	Mumbai	85
4	Diana	Mumbai	88
6	Frank	Mumbai	65
9	Ivan	Mumbai	89

Q5) Write a MySQL query to display students whose marks are greater than 80.

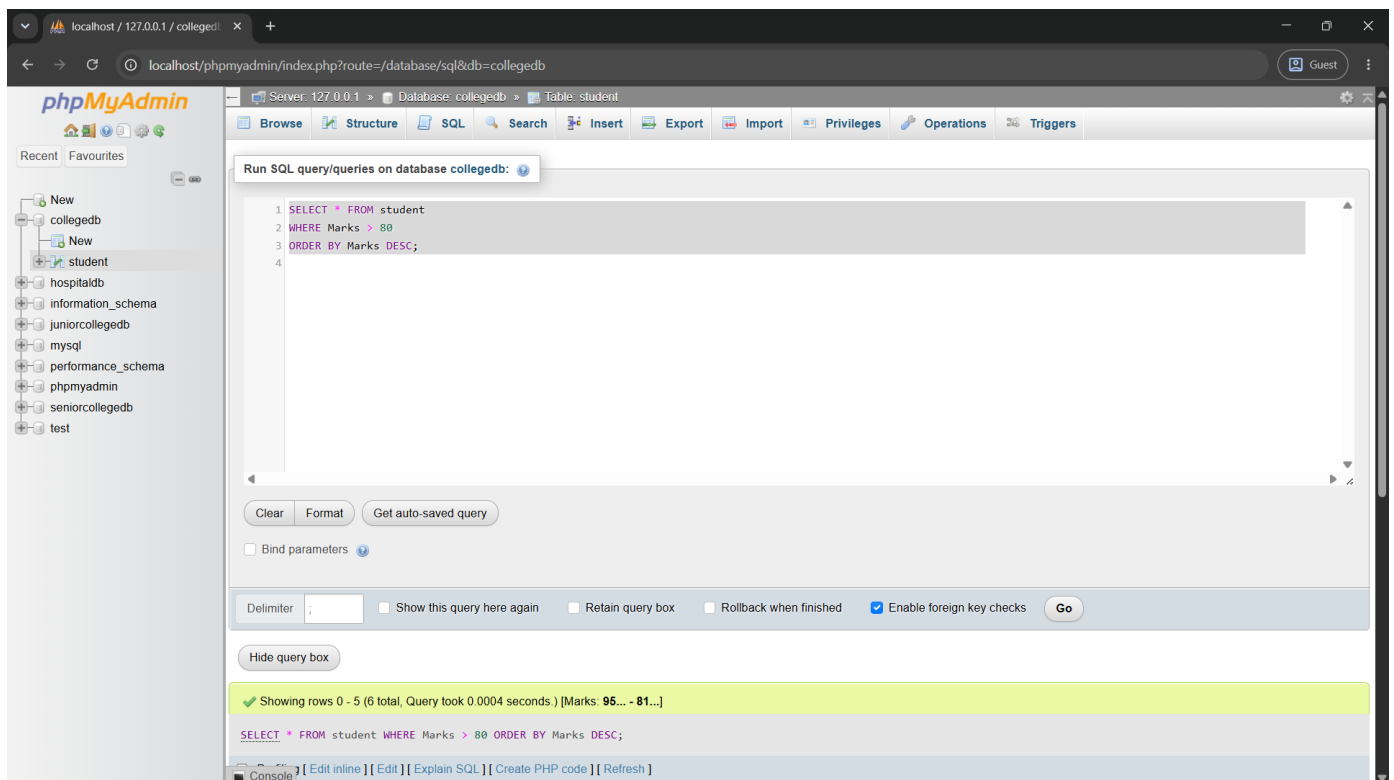
The output should be sorted by marks in descending order.

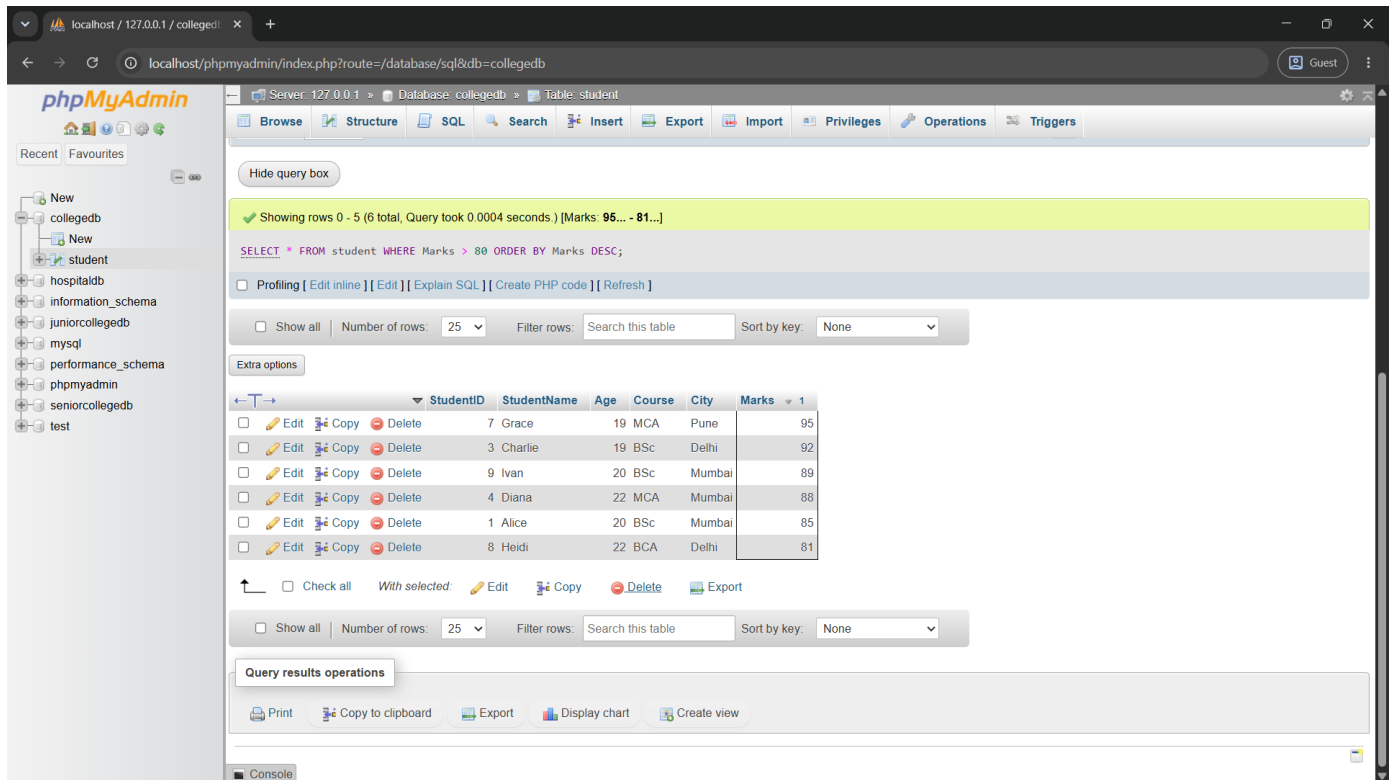
QUERY:

SELECT * FROM student

WHERE Marks > 80

ORDER BY Marks DESC;





Q6) Write a MySQL query to display the top 3 students based on marks. The query should:

1. Display StudentName and Marks.
2. Sort marks from highest to lowest.
3. Display only 3 records using LIMIT.

QUERY:

SELECT StudentName,Marks FROM student

ORDER by Marks DESC

LIMIT 3;

localhost / 127.0.0.1 / collegedb

localhost/phpmyadmin/index.php?route=/database/sql&db=collegedb

phpMyAdmin

Recent Favourites

- New
- collegedb
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- seniorcollegedb
- test

Server: 127.0.0.1 > Database: collegedb > Table: student

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Run SQL query/queries on database collegedb:

```
1 SELECT StudentName,Marks FROM student
2 ORDER by Marks DESC
3 LIMIT 3;
4
```

Clear Format Get auto-saved query

☐ Bind parameters

Delimiter ; ☐ Show this query here again ☐ Retain query box ☐ Rollback when finished ☒ Enable foreign key checks Go

Hide query box

Showing rows 0 - 2 (3 total, Query took 0.0002 seconds.) [Marks: 95... - 89...]

```
SELECT StudentName,Marks FROM student ORDER by Marks DESC LIMIT 3;
```

Console [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

localhost / 127.0.0.1 / collegedb

localhost/phpmyadmin/index.php?route=/database/sql&db=collegedb

phpMyAdmin

Recent Favourites

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Delimiter ; ☐ Show this query here again ☐ Retain query box ☐ Rollback when finished ☒ Enable foreign key checks Go

Hide query box

Showing rows 0 - 2 (3 total, Query took 0.0002 seconds.) [Marks: 95... - 89...]

```
SELECT StudentName,Marks FROM student ORDER by Marks DESC LIMIT 3;
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Extra options

	StudentName	Marks
☐ Edit Copy Delete	Grace	95
☐ Edit Copy Delete	Charlie	92
☐ Edit Copy Delete	Ivan	89

Check all With selected: Edit Copy Delete Export

Query results operations

Print Copy to clipboard Export Display chart Create view

Console